Reading a research claim

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

THE CONNECTED WEEK

Unpack the outcome. Keep the denominator. Audit the headline.

Read descriptive research claims without confusing absolute/relative change or adding unmeasured learner effects.

Keep one habit: Claim contract → eight-case arithmetic → ten-case doubling picture → unit/scope probes → exact-count audit → independent local-median claim.

MAIN LESSON · SECTION 1

Read the claim through its outcome, denominator and conditions

> A headline can extend beyond the comparison its table actually supports.

Learning goals

I can explain the **mechanism** and terminology of reading a research **claim**.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Retrieve bounded questions and fixed scoring

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from A question that **evidence** can answer. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Prerequisite and starting response

- Recall A question that **evidence** can answer. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a **claim**.

> **My starting explanation**

End goal

Explain reading a research **claim** using the records and boundaries of this fictional investigation.

▼ **Earlier wording, alternatives and picture transcripts**

Your learning target

Remember before reading

This week you will practise:

Explain the **mechanism** and terminology of reading a research **claim**

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Retrieve what you already know

This week’s end goal

MAIN LESSON · SECTION 2

Separate the headline, measured variable and evidence gaps

> A descriptive improvement is narrower than a universal benefit.

Begin with “better at what?”

A headline says “B is better.” Better at which measured outcome, compared with what, and on how many cases? Read the method and result together. In the main fictional report, supported-answer counts rise from 5/8 to 6/8 on the same eight constructed cases. That is a higher observed rate for this outcome. It is not a result about every task, each learner or an unmeasured cause.

Ask for the scoring and conditions that make the count interpretable.

Define reading a research claim

Reading a **claim** requires its method, data, comparison and uncertainty. Titles and abstracts may omit conditions crucial to interpretation.

Complete denominator and measurement contract

Claim-audit item	Complete supplied fictional evidence
Main dataset/unit	Eight fixed constructed answer cases; case is the reported unit
Main baseline A	5 supported answers / 8 cases
Main method B	6 supported answers / same 8 cases

Claim-audit item	Complete supplied fictional evidence
Main outcome	Supplied supported-answer flag; do not relabel it learner improvement or overall usefulness
Main descriptive change	62.5% to 75%; +12.5 percentage points; ratio 1.2
Missing method details	Exact support rubric, versions, evaluation conditions and sampling/uncertainty analysis not supplied by the headline
Separate picture	Correct-answer count 2/10 to 4/10 on same ten cases with fixed scoring
Picture descriptive change	20% to 40%; +20 percentage points; rate ratio 2, relative increase 100%
Limits	No population interval, significance result, causal effect or individual-learning evidence supplied
Evidence status	All counts and any added possible paired records are fictional examples, not a real paper or new experiment

Name supplied and missing evidence

List the given facts and one fact that is not supplied.

▼ **Earlier wording, alternatives and picture transcripts**

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

MAIN LESSON · SECTION 3

Trace counts, rates and the exact scope of “better”

> Keep percentage points separate from relative percent change.

Original claim-versus-evidence reading procedure

Separate the **claim** from supplied **evidence**. Record dataset, unit, baseline, metric and limitations. Ask whether the statement extends beyond those **observation**s.

Complete main arithmetic and inference ledger

Main fictional comparison	Baseline A	Method B	Supported descriptive statement
Supported count / same cases	5 / 8	6 / 8	One more supported answer in these counts
Observed percentage	62.5%	75%	+12.5 percentage points
Rate ratio	Reference	(6/8) / (5/8) = 1.2	20% relative increase in this measured rate
Case-level transitions	Not supplied	Not supplied	Totals do not show who gained or regressed
Broader inference	No sampling/uncertainty analysis	No general task or learner outcomes	Do not claim universal superiority or causal effect

State the missing method and uncertainty evidence

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Explain the visual

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ Earlier wording, alternatives and picture transcripts

Reading a research claim — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
A fictional report says method B is better after 6/8 supported answers versus baseline 5/8 on the same constructed cases.	Separate the claim from supplied evidence. Record dataset, unit, baseline, metric and limitations. Ask whether the statement extends beyond those observations.	B has 75% versus 62.5%, a 12.5-percentage-point difference on eight cases. Better in this measured outcome does not establish broader tasks or uncertainty.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Connect the picture and the explanation

Predict a headline that suddenly claims every learner improves

> Changing the words does not add an unmeasured learner outcome.

Current task and response

- A fictional report says method B is better after 6/8 supported answers versus baseline 5/8 on the same constructed cases. Show the intermediate steps and give the conclusion with its scope.

> **Keep the procedure fixed. Explain what changed in the inputs and result.**

Predict the stronger learner headline

- Change: The headline changes to "every learner improves twice as much".

▼ **Check after predicting**

Prediction: That learner **claim** is unsupported by these answer counts.

Reason: The outcome and experimental unit have changed in the wording without new **evidence**.

▼ **Earlier wording, alternatives and picture transcripts**

Compare a familiar case with a changed case

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Read “twice the rate” with its ten-case denominator

> A correct ratio still needs a bounded interpretation.

Main eight-case claim and checked scope

A fictional report says method B is better after 6/8 supported answers versus baseline 5/8 on the same constructed cases.

B has 75% versus 62.5%, a 12.5-percentage-point difference on eight cases. Better in this measured outcome does not establish broader tasks or uncertainty.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Separate prompt counts and full calculations

Read a **claim** with its denominator. A fictional paper headline says a prompt is twice as effective. Its table reports four correct cases instead of two.

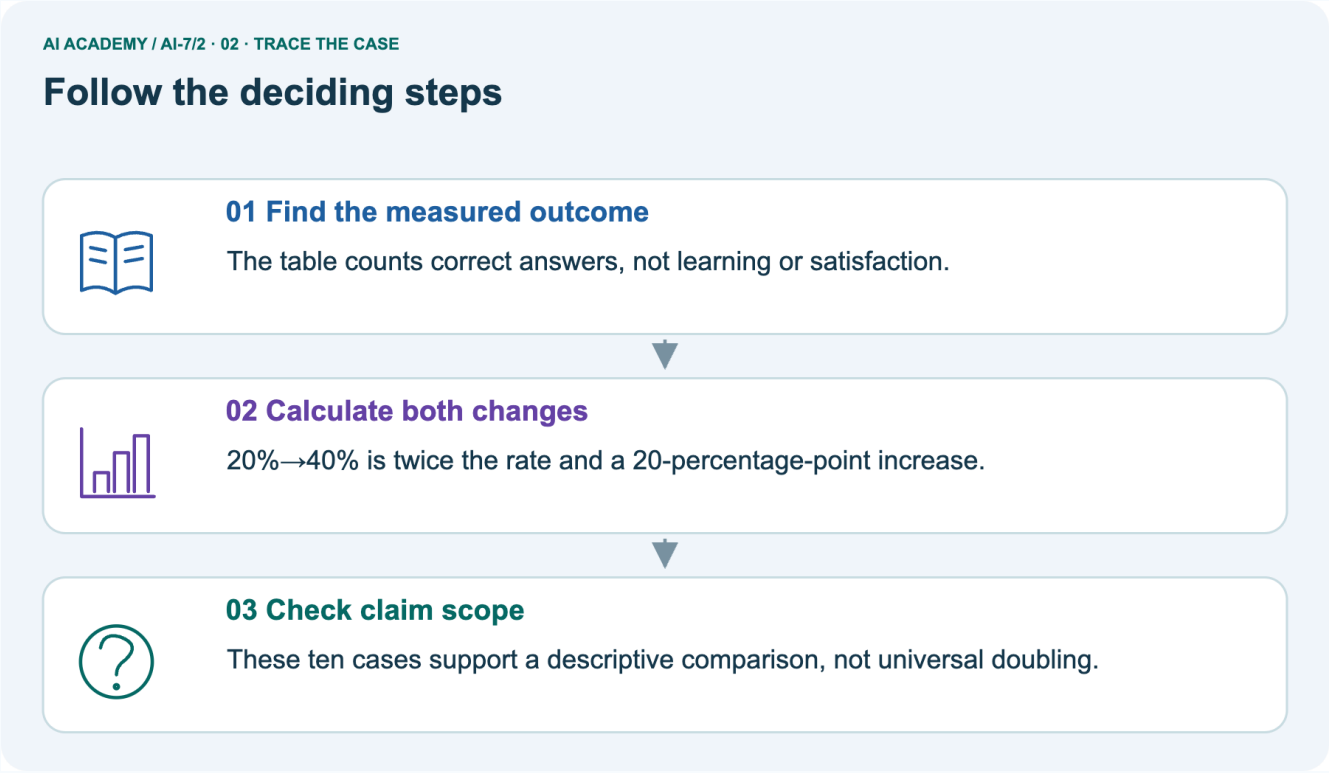
Baseline: 2/10 correct. New prompt: 4/10 on the same ten cases. Scoring is fixed.

The picture is a separate fictional prompt comparison: baseline 2/10 correct versus new 4/10, both under the same ten-case scoring. Rates are 20% and 40%. The latter is twice the former, a 100% relative increase, but the absolute change is 20 percentage points and the new prompt still fails six of ten cases. None of these quantities measures each learner’s improvement.

The changed headline “every learner improves twice as much” changes both the unit and outcome without supplying learner data. No significance, population interval or causal study is provided; do not replace that gap with either a claim of statistical significance or a claim that no real effect exists.

Separate picture quantity	Calculation	Meaning and limit
Baseline correctness	2/10 × 100 = 20%	Observed correct-answer rate
New correctness	4/10 × 100 = 40%	Observed rate; six cases still incorrect
Absolute rate change	40% – 20% = 20 percentage points	Not a 20% relative increase
Rate ratio	40% / 20% = 2	Twice the observed rate
Relative increase	(40% – 20%)/20% = 100%	Not twice every learner's improvement
Learner outcome	No learner measurements supplied	Cannot infer individual learning effects

Trace outcome, absolute/relative change and claim limits



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Find the measured outcome	The table counts correct answers, not learning or satisfaction.
2. Calculate both changes	20%→40% is twice the rate and a 20-percentage-point increase.
3. Check claim scope	These ten cases support a descriptive comparison, not universal doubling.

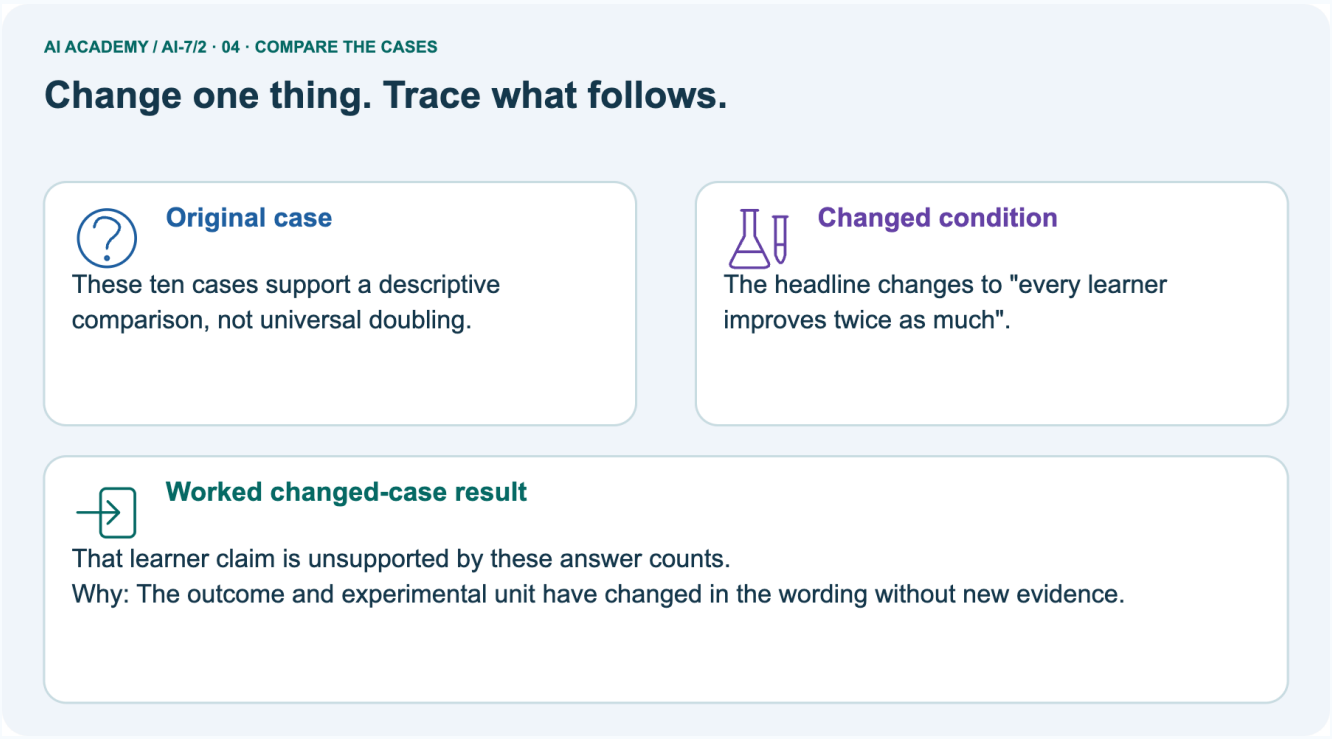
Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

A research **claim** compresses a method and result. Unpack the measured variable, denominator and comparison before accepting the headline.

Predict “every learner improves twice as much”

- Change one thing: The headline changes to "every learner improves twice as much". Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain missing uncertainty and learner evidence

- Small case counts leave substantial uncertainty; no significance or causal effect is supplied.
- Read the labelled picture: Calculate the baseline and new rates for 2/10 and 4/10. Explain both twice the rate and the percentage-point change, and say why neither measures every learner's improvement.

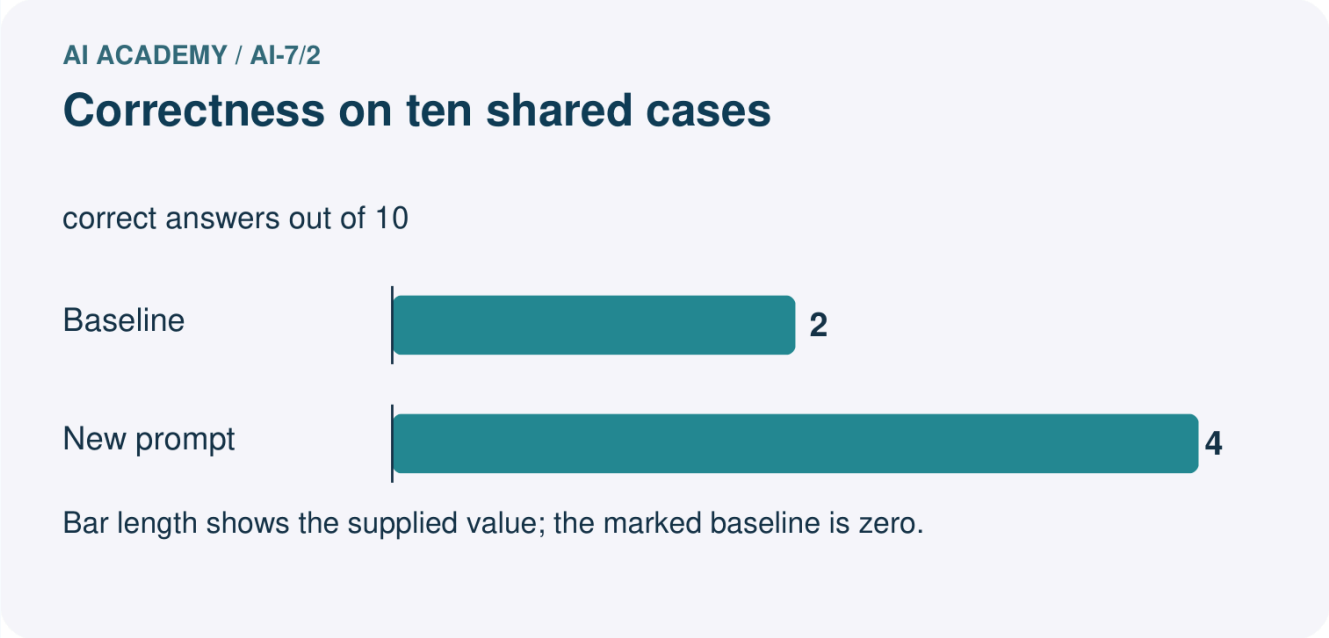
▼ Earlier wording, alternatives and picture transcripts

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Reading a research claim

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-7/2

Reading a research claim

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Find the measured outcome

The table counts correct answers, not learning or satisfaction.

2. Calculate both changes

20% -> 40% is twice the rate and a 20-percentage-point increase.

3. Check claim scope

These ten cases support a descriptive comparison, not universal doubling.

CHANGE ONE THING

The headline changes to "every learner improves twice as much".

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Audit the measured variable and the reach of the conclusion

> Keep missing evidence visible instead of accepting the headline's extra meaning.

Core claim qualification and failure analysis

A numerical improvement can be real in the sample while the broader **claim** remains unsupported.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Complete rate, unit and inference probes

Claim-audit probe	Expected interpretation
Count improves with same denominator	Compute a descriptive rate difference under that outcome
"Better" omits metric or baseline	Ask what was measured and compared
Percent change confused with percentage points	Write both starting and ending rates before calculating
Rate ratio with zero baseline	Undefined; do not invent a finite doubling ratio
Same aggregate counts, different paired records	Individual gains/losses are not identified by totals
Headline names learners but table counts answers	Unit/outcome mismatch
No uncertainty analysis supplied	No significance or population-precision claim established
Matched cases alone	Useful comparability condition, not automatic causal identification

Claim-audit probe	Expected interpretation
One small fictional report	No universal task, learner or deployment conclusion

Guided repair and explanation

- A report omits this qualification: A numerical improvement can be real in the sample while the broader **claim** remains unsupported.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

State one repair and the checks you would rerun.

▼ Earlier wording, alternatives and picture transcripts

Locate the earliest unsupported step

Practise rewriting a claim at the strength its evidence supports

> Retain the result while removing the unsupported generalization.

Current explanation and guided check

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.**

> **Explain your prediction to a partner before comparing results.**

Supported W2 trace

Use workbook W2 for the connected example “Reading a research **claim**”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Individual scope check

- Use the worked case for Reading a research **claim**. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Picture response

- Calculate the baseline and new rates for 2/10 and 4/10. Explain both twice the rate and the percentage-point change, and say why neither measures every learner’s improvement.

> **My explanation using a pictured step and the given facts**
>
> _____
>
> _____
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> _____
>
> _____

Reserve the selected local-median universal-speed claim

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

Practise with less help: Trace the worked case

Check your reasoning

Explain the picture

Run the complete count-to-claim arithmetic investigation

> Inspect which conclusions the aggregates identify and which they leave unknown.

Current “B is better” claim investigation

A fictional report says method B is better after 6/8 supported answers versus baseline 5/8 on the same constructed cases. Show the intermediate steps and give the conclusion with its scope.

Runnable exact-count and possible-record audit

```
"""Exact arithmetic on fictional claim counts, not a study or statistical test."""
from fractions import Fraction

def proportion(passed,total):
 if type(passed) is not int or type(total) is not int or total<=0 or not 0<=passed<=total:
  raise ValueError('valid exact count and positive denominator required')
 return Fraction(passed,total)
def compare_counts(a_pass,a_total,b_pass,b_total):
 a=proportion(a_pass,a_total);b=proportion(b_pass,b_total)
 return {'baseline_rate':a,'new_rate':b,'difference_percentage_points':100*(b-a),
         'rate_ratio':b/a if a else None,'relative_increase_percent':100*(b-a)/a if a else
None}
main=compare_counts(5,8,6,8)
picture=compare_counts(2,10,4,10)
assert main['baseline_rate']==Fraction(5,8) and main['new_rate']==Fraction(3,4)
assert main['difference_percentage_points']==Fraction(25,2) and
main['rate_ratio']==Fraction(6,5)
assert picture['difference_percentage_points']==20 and picture['rate_ratio']==2 and
picture['relative_increase_percent']==100
# Added possible paired records: both fit the same main aggregate counts.
a=[True,True,True,True,True,False,False,False]
b_one_gain=[True,True,True,True,True,True,False,False]
b_gain_and_loss=[True,True,True,True,False,True,True,False]
def paired_changes(a,b):
 if not a or len(a)!=len(b) or any(type(v) is not bool for v in a+b):raise
ValueError('matched Boolean fixtures required')
 return {'gains':sum(not x and y for x,y in zip(a,b)),'losses':sum(x and not y for x,y in
zip(a,b))}
pattern_one=paired_changes(a,b_one_gain);pattern_two=paired_changes(a,b_gain_and_loss)
assert sum(a)==5 and sum(b_one_gain)==sum(b_gain_and_loss)==6
assert pattern_one=={'gains':1,'losses':0} and pattern_two=={'gains':2,'losses':1}
```

```
assert compare_counts(0,10,1,10)['rate_ratio'] is None
print('Main supported-answer rates:',main,'62.5 percent versus 75 percent on eight
supplied cases.')
print('Separate picture correctness rates:',picture,'20 percent versus 40 percent on ten
supplied cases.')
print('Two added possible paired patterns with identical main
totals:',pattern_one,pattern_two)
print('Aggregates do not reveal every case outcome, individual learning, causal effect or
population uncertainty. These are constructed counts/possible records, not a real paper,
model run or statistical significance test.')
```

Expected arithmetic and evidence-status limits

1. Predict main rates and percentage-point change before running or tracing, then separately compute the picture's doubling ratio and 20-point absolute change. The exact-fraction code also constructs two possible paired patterns with identical main totals: one gain/no losses versus two gains/one loss. Those are possible illustrations, not recovered case records from the report. A zero baseline has no finite rate ratio.

   No actual paper, model, participant, statistical significance test or causal estimate is involved. Keep the independent median-time claim for the first individual response.

Evidence ledger and reflection

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or "not run"	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ **Earlier wording, alternatives and picture transcripts**

Predict, investigate and preserve the evidence

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: four local fixtures and a universal-speed claim

> Keep the observed median difference while auditing its scope.

Complete selected independent task

Report **claim**s universal speed after median 2 seconds versus 3 on four local fixtures. Audit scope.

First attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ Earlier wording, alternatives and picture transcripts

Independent case: Apply a new case independently

Attempt this without hints

Report **claim**s universal speed after median 2 seconds versus 3 on four local fixtures. Audit scope.

> **My independent procedure, result and limitation**
>
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> | |

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

MAIN LESSON · SECTION 11

Save claim, method, data, comparison and missing uncertainty together

> Distinguish a faithful summary from a stronger headline.

Project task and artifact

Add a labelled reading a research **claim evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of reading a research **claim**.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Actual arithmetic run or paper trace and limitations

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the **claim** to its records, units and assumptions so another investigator can challenge or reproduce it.

▼ **Earlier wording, alternatives and picture transcripts**

Evidence for your project

MAIN LESSON · SECTION 16

Explain one measured comparison and one unsupported extension

> A narrow valid result need not justify a broad headline.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of reading a research **claim**

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Use research vocabulary precisely**

REFERENCE · SECTION 5

Reference · Use research vocabulary precisely

> Connect each term to the question, comparison, measurement or evidence.

Vocabulary and your explanation

Word	Meaning
Claim	A statement offered as a conclusion
Evidence scope	The conditions represented by observations
Extrapolation	Extending beyond measured conditions

Use two terms to explain a step in this week's example.

▼ Extra challenge · Explain a stronger case

REFERENCE · SECTION 10

Extra challenge · Explain a stronger case

> Choose the next challenge after the main route.

Explain,repair and extend

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

- A report omits this qualification: A numerical improvement can be real in the sample while the broader **claim** remains unsupported.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

Explain → apply → challenge

> **Explain which evidence supports your conclusion and which claim remains unproved.**

▼ Reference · Check the research reasoning and its limits

REFERENCE · SECTION 12

Reference · Check the research reasoning and its limits

> Use after practice; reserve the selected independent case for its first attempt.

Current explanation and response

- Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

  For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

My limited claim and a useful next check

▼ **Extra practice · Revisit the complete trace**

REFERENCE · SECTION 13

Extra practice · Revisit the complete trace

> Predict before reading the supplied steps.

Complete trace and evidence

Complete evidence record

Reading a research claim — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
A fictional report says method B is better after 6/8 supported answers versus baseline 5/8 on the same constructed cases.	Separate the claim from supplied evidence. Record dataset, unit, baseline, metric and limitations. Ask whether the statement extends beyond those observations.	B has 75% versus 62.5%, a 12.5-percentage-point difference on eight cases. Better in this measured outcome does not establish broader tasks or uncertainty.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

▼ Reference · The earlier route checklist

REFERENCE · SECTION 15

Reference · The earlier route checklist

> Use one main route rather than repeating the lesson.

Targets,route and first prediction

Start with this question: Read a **claim** with its denominator. A fictional paper headline says a prompt is twice as effective. Its table reports four correct cases instead of two.

By the end, I can	How I will show it
Explain the mechanism and terminology of reading a research claim	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Extra practice · Predict,check and repair

REFERENCE · SECTION 17

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Read a **claim** with its denominator. A fictional paper headline says a prompt is twice as effective. Its table reports four correct cases instead of two.
- Which labelled step matches this detail? These ten cases support a descriptive comparison, not universal doubling.
- 1. Check **claim** scope
- 2. Calculate both changes
- 3. Find the measured outcome
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Check **claim** scope These ten cases support a descriptive comparison, not universal doubling. A research **claim** compresses a method and result.

Unpack the measured variable, denominator and comparison before accepting the headline.

Changed-case question and optional hint

- Changed-case practice: The headline changes to "every learner improves twice as much". Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. This one example guarantees correct results for every future **input**.
- 3. That learner **claim** is unsupported by these answer counts.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: That learner **claim** is unsupported by these answer counts. The outcome and experimental unit have changed in the wording without new **evidence**.

Repair and retain the record

Review A question that **evidence** can answer (ai-7/1). Goal: Explain the **mechanism** and terminology of a question that **evidence** can answer. Short practice: Which proposed diagram question has a measurable outcome?

Name its versions, six-case denominator and fixed scoring, then explain why giving only B easier tasks spoils the comparison.. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work
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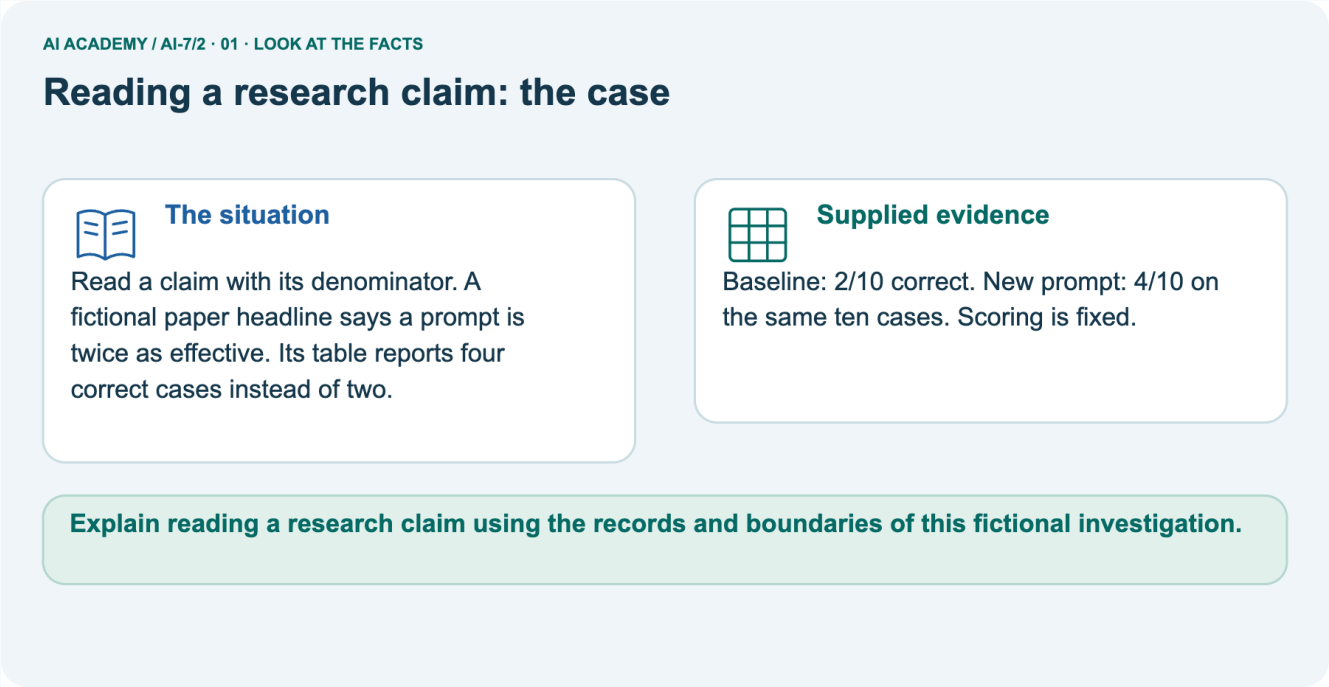
<details>
<summary>Reference · Six-picture concept journey</summary>
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REFERENCE · SECTION 18

Reference · Six-picture concept journey

> Choose the picture that supports the current explanation.

01. Reading a research claim: the case



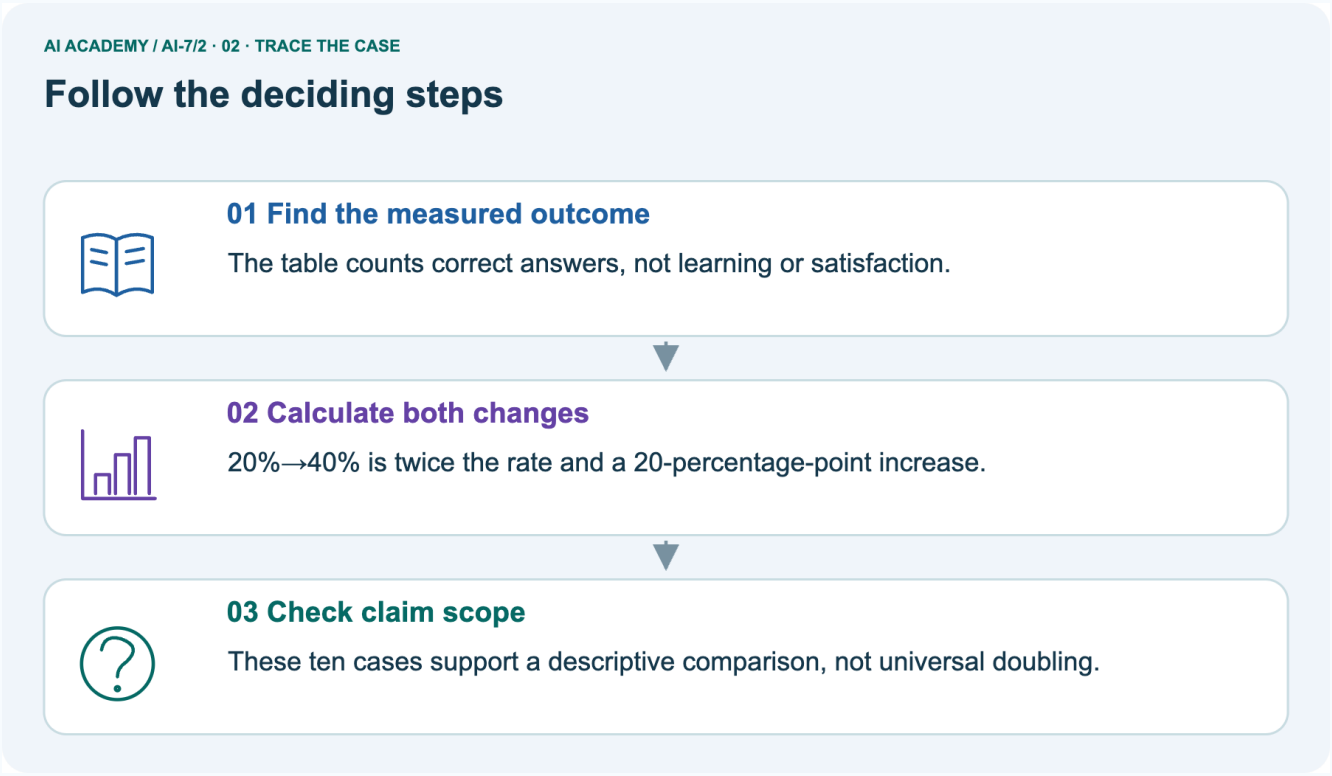
Follow the labelled picture with the matching explanation. Open the vector diagram

Read a **claim** with its denominator. A fictional paper headline says a prompt is twice as effective. Its table reports four correct cases instead of two. Supplied facts: Baseline: 2/10 correct.

New prompt: 4/10 on the same ten cases. Scoring is fixed.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

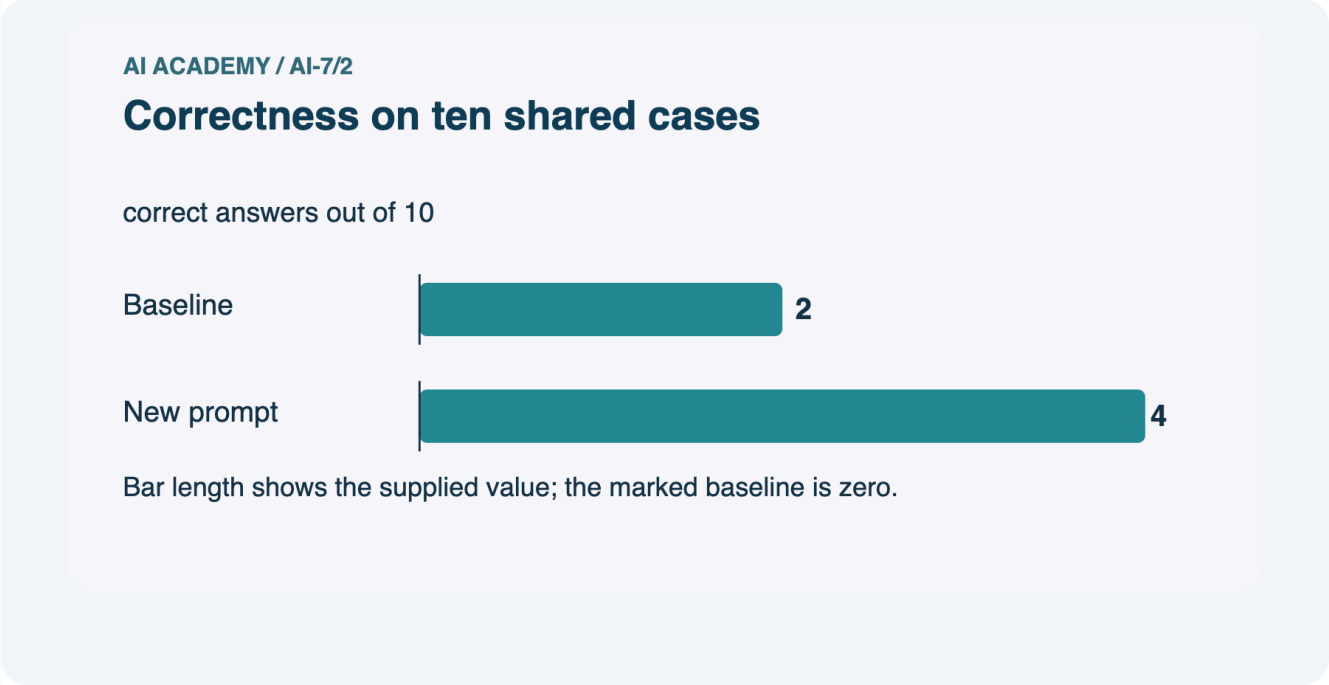


Follow the labelled picture with the matching explanation. Open the vector diagram

Find the measured outcome: The table counts correct answers, not learning or satisfaction.
→ Calculate both changes: 20%→40% is twice the rate and a 20-percentage-point increase.
→ Check **claim** scope: These ten cases support a descriptive comparison, not universal doubling.

Point to each numbered stage and explain which recorded **input** it uses.

03. Correctness on ten shared cases

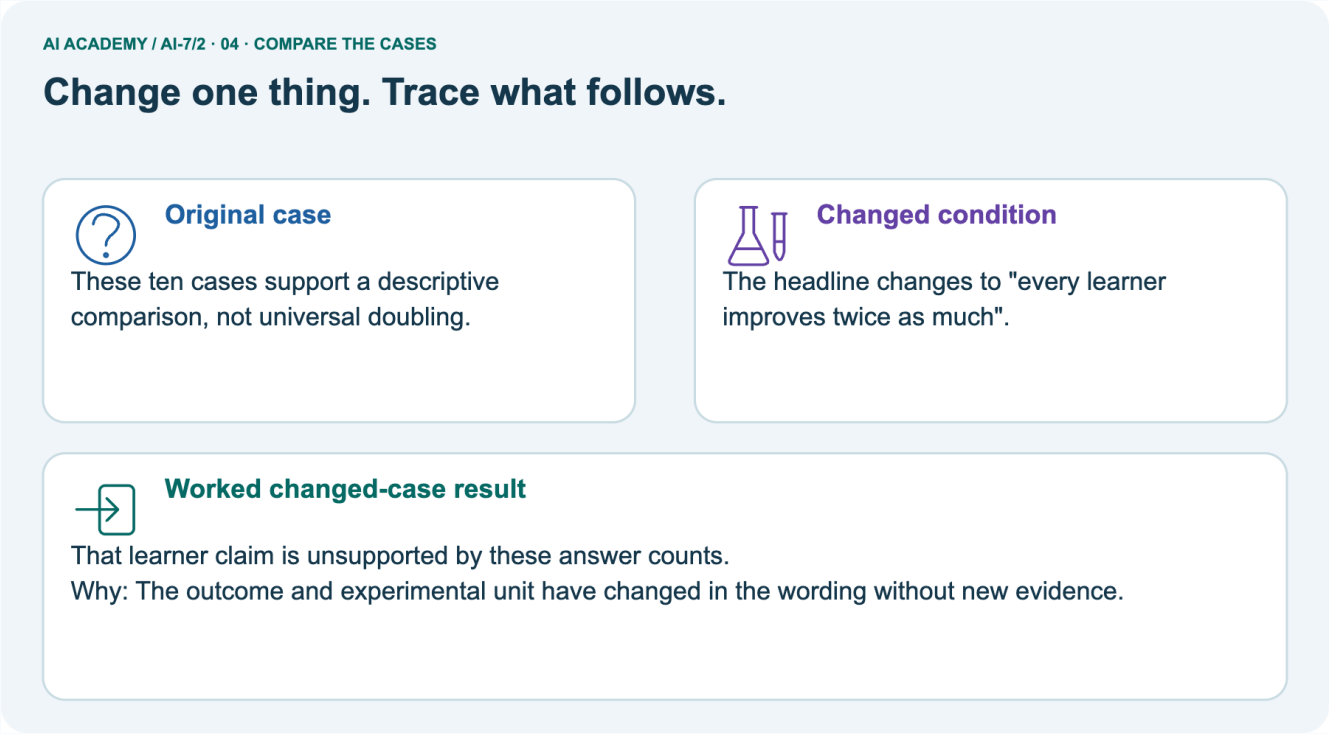


Follow the labelled picture with the matching explanation. Open the vector diagram

Correctness on ten shared cases. Units: correct answers out of 10. Baseline: 2. New prompt: 4.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



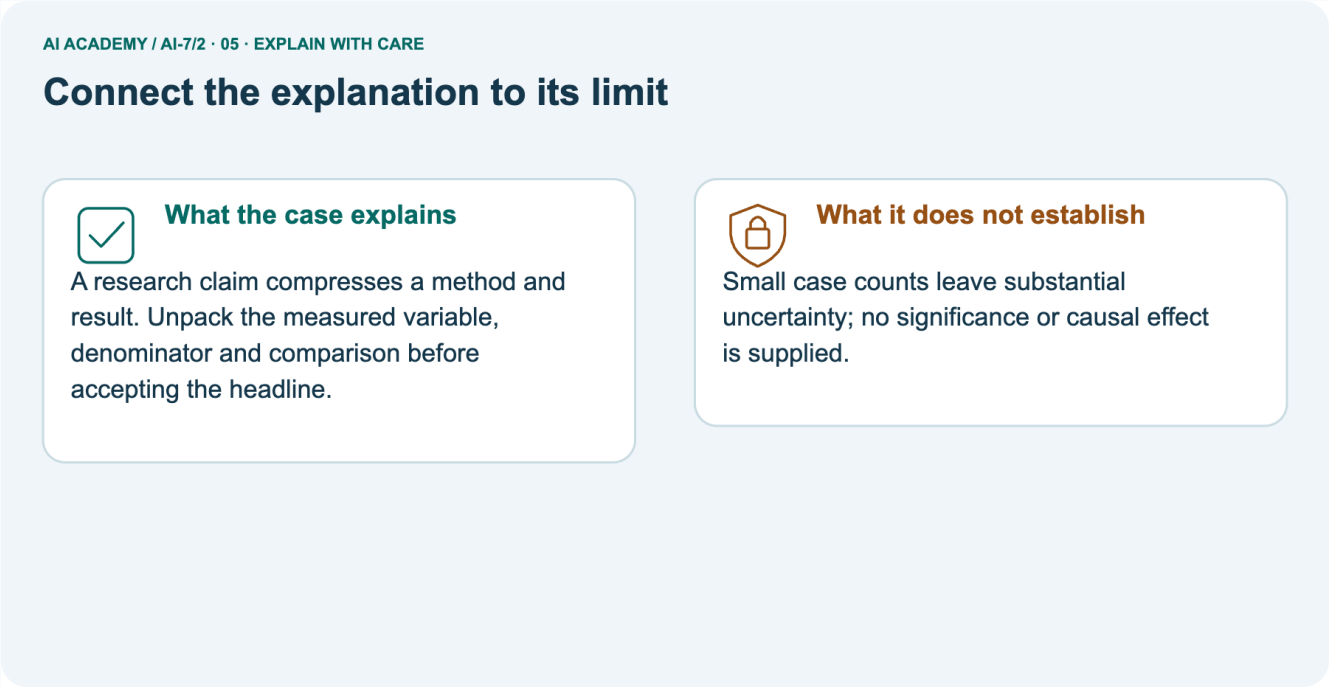
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: These ten cases support a descriptive comparison, not universal doubling.
Changed condition: The headline changes to "every learner improves twice as much".

Expected result: That learner **claim** is unsupported by these answer counts. Reason: The outcome and experimental unit have changed in the wording without new **evidence**.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



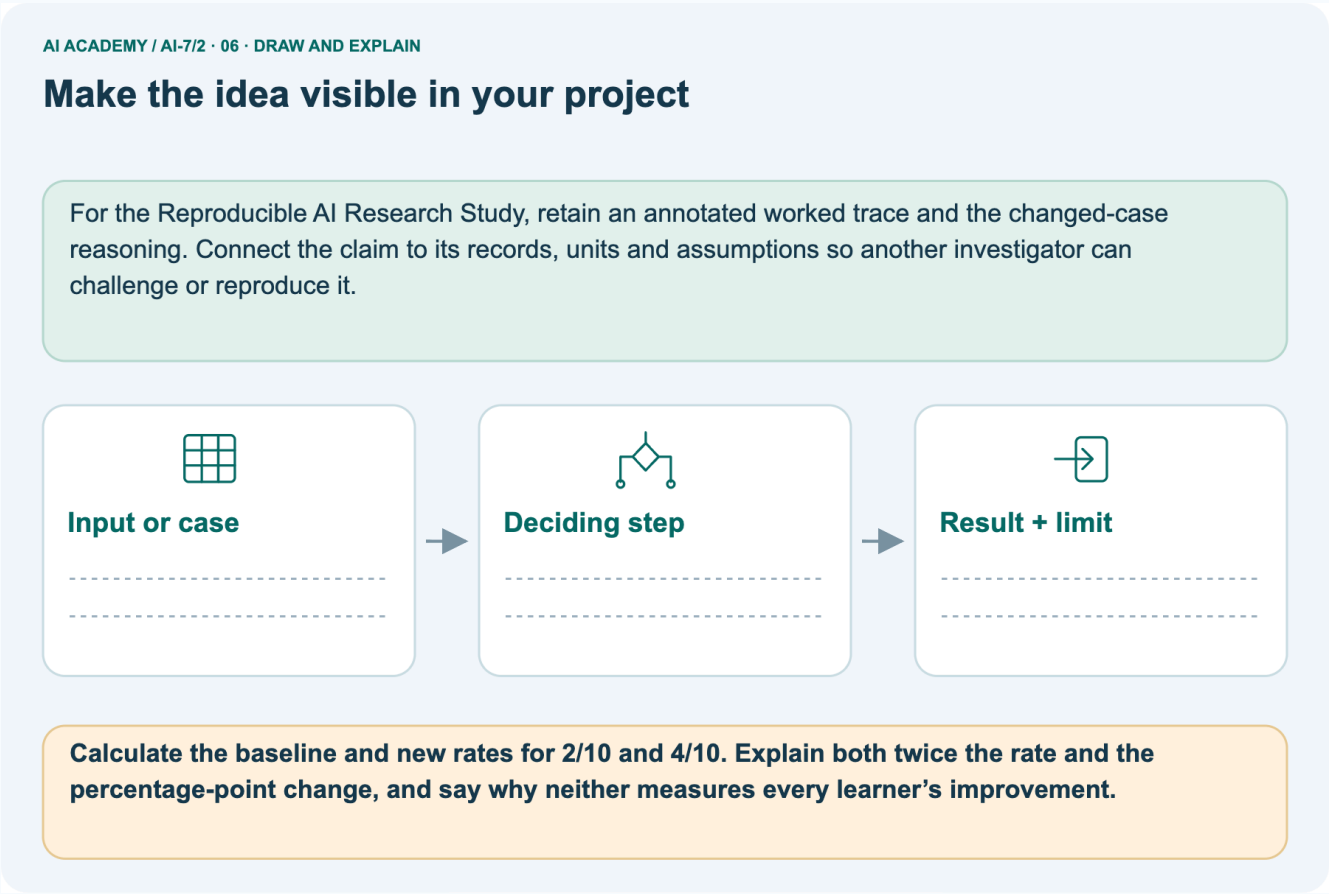
Follow the labelled picture with the matching explanation. Open the vector diagram

A research **claim** compresses a method and result. Unpack the measured variable, denominator and comparison before accepting the headline.

Limit: Small case counts leave substantial uncertainty; no significance or causal effect is supplied.

Explain the case, then point to a limit on the **claim**.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the **claim** to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Calculate the baseline and new rates for 2/10 and 4/10.

Explain both twice the rate and the percentage-point change, and say why neither measures every learner's improvement.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Reading a research claim: the case

02. Follow the deciding steps

03. Correctness on ten shared cases

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.